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Phase Optimization in Reconfigurable Intelligent Surfaces for Signal Enhancement

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Problem Statement

Wireless communication systems are affected by multipath propagation, where transmitted signals reflect from obstacles such as buildings and walls before reaching the receiver. These signals arrive with different phase shifts, often causing destructive interference and reducing signal strength and quality.

Traditional approaches such as increasing transmission power or adding hardware are inefficient and costly. Moreover, conventional systems treat the wireless environment as passive and uncontrollable.

Therefore, there is a need for an intelligent and energy-efficient solution to control signal reflections. Reconfigurable Intelligent Surfaces (RIS) enable programmable control over these reflections. This work focuses on using RIS-based phase optimization to align signals constructively and enhance received signal strength.

Objectives

- To evaluate and compare the received signal power for three scenarios: without RIS, with randomly configured RIS, and with optimized RIS — across varying SNR levels (0 to 20 dB) with Monte Carlo averaging over 500 iterations. To analyze how additive noise affects received power in each configuration.

- To analyze how increasing the number of RIS elements (N = 10, 20, 50, 100) affects the average received signal power under optimized phase configuration, using 200 Monte Carlo iterations per case.

- To measure and illustrate the SNR gain (in dB) achieved by optimized RIS phase configuration over the no-RIS baseline, computed as the difference in received power across all tested SNR values.

- To demonstrate how phase optimization aligns the reflected signals from 20 RIS elements by visualizing the phase distribution of each element's contribution — comparing random phase assignment versus optimized phase alignment — to show the conversion of destructive interference into constructive interference.

Introduction

Reconfigurable Intelligent Surfaces (RIS) represent a transformative approach in modern wireless communication systems, where the propagation environment itself becomes controllable. An RIS is composed of a large number of passive reflecting elements, each capable of independently adjusting the phase of an incoming electromagnetic wave. By coordinating these elements, it becomes possible to control how signals are reflected and directed within a communication system.

In traditional wireless communication, signals travel from the transmitter to the receiver through multiple paths due to reflections, scattering, and diffraction. These multiple paths introduce different phase shifts, which can result in destructive interference when signals combine at the receiver. This leads to reduced signal strength, poor quality of service, and unreliable communication.

RIS overcomes this limitation by intelligently controlling these reflections. By applying appropriate phase shifts at each reflecting element, the signals can be aligned in such a way that they add constructively at the receiver. This results in improved signal strength, better coverage, and enhanced communication reliability.

Unlike conventional solutions that rely on increasing transmission power or adding more hardware components, RIS offers a low-power, cost-effective, and scalable solution. It shifts the focus from optimizing the transmitter and receiver to optimizing the environment itself. This makes RIS a key enabling technology for future communication systems, particularly in the development of next-generation networks such as 6G.

Implementation

The proposed system is implemented using MATLAB to simulate a wireless communication scenario incorporating Reconfigurable Intelligent Surfaces. The simulation evaluates signal transmission across SNR values from 0 to 20 dB, using Monte Carlo averaging over 500 iterations to ensure statistically reliable results.

Three scenarios are considered: without RIS, with randomly phase-shifted RIS, and with optimized RIS. The wireless channels are modeled using complex Gaussian random variables to represent Rayleigh fading conditions. For the optimized case, each RIS element's phase shift is set to the negative of the combined transmitter-to-RIS and RIS-to-receiver channel phase, ensuring maximum constructive interference at the receiver.

Performance is evaluated through four analyses: received signal power comparison across the three scenarios, effect of RIS array size by varying elements from $N = 10$ to 100, gain achieved by optimized RIS over the no-RIS baseline, and phase distribution visualization to illustrate the conversion of destructive interference into constructive interference.

Results and Discussion

The simulation was executed in MATLAB and produced four key results, each analyzed below.

SNR Comparison: RIS vs No RIS

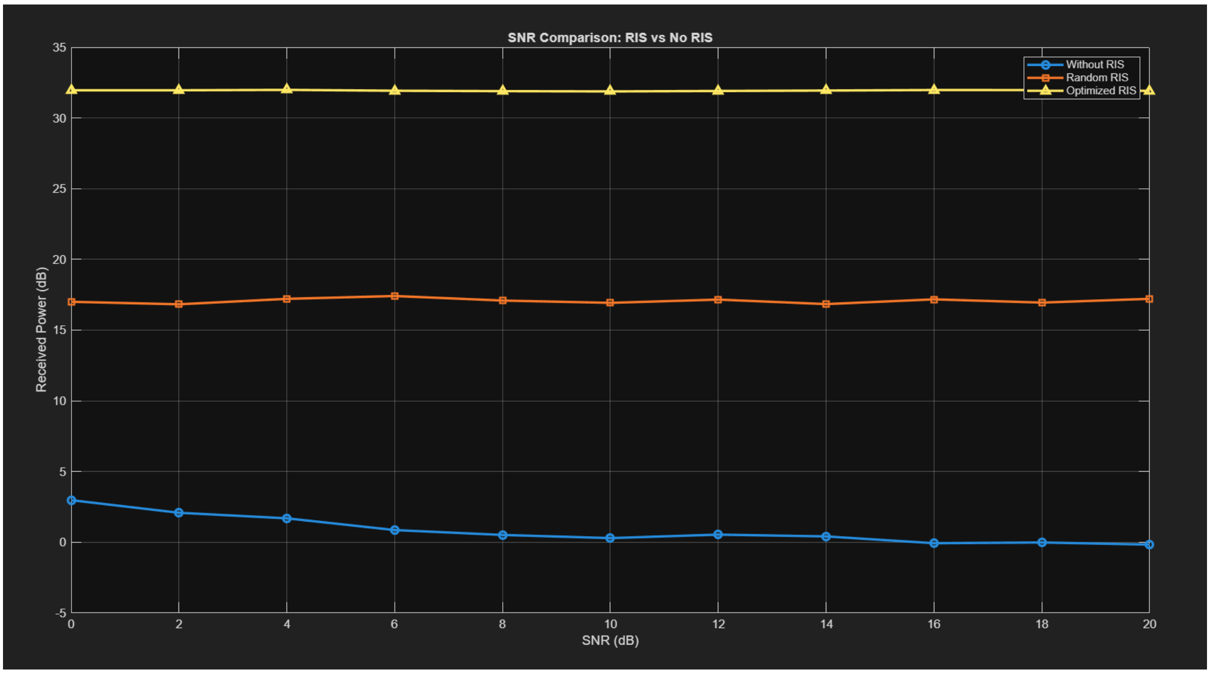

Figure 1: Received Power vs SNR for three scenarios: Without RIS, Random RIS, and Optimized RIS (N=50, 500 Monte Carlo iterations)

Figure 1 compares the received signal power across SNR values ranging from 0 to 20 dB for all three scenarios. The optimized RIS consistently achieves approximately 32 dB of received power, remaining nearly flat across all SNR values, demonstrating that phase optimization dominates over noise effects. The random RIS settles around 17 dB, offering moderate improvement due to partial constructive interference. The no-RIS baseline starts near 3 dB at 0 dB SNR and gradually decreases toward 0 dB as noise power increases, confirming the adverse effect of noise on the direct channel alone. The clear separation between all three curves validates the effectiveness of RIS phase optimization.

Effect of RIS Size on Signal Strength

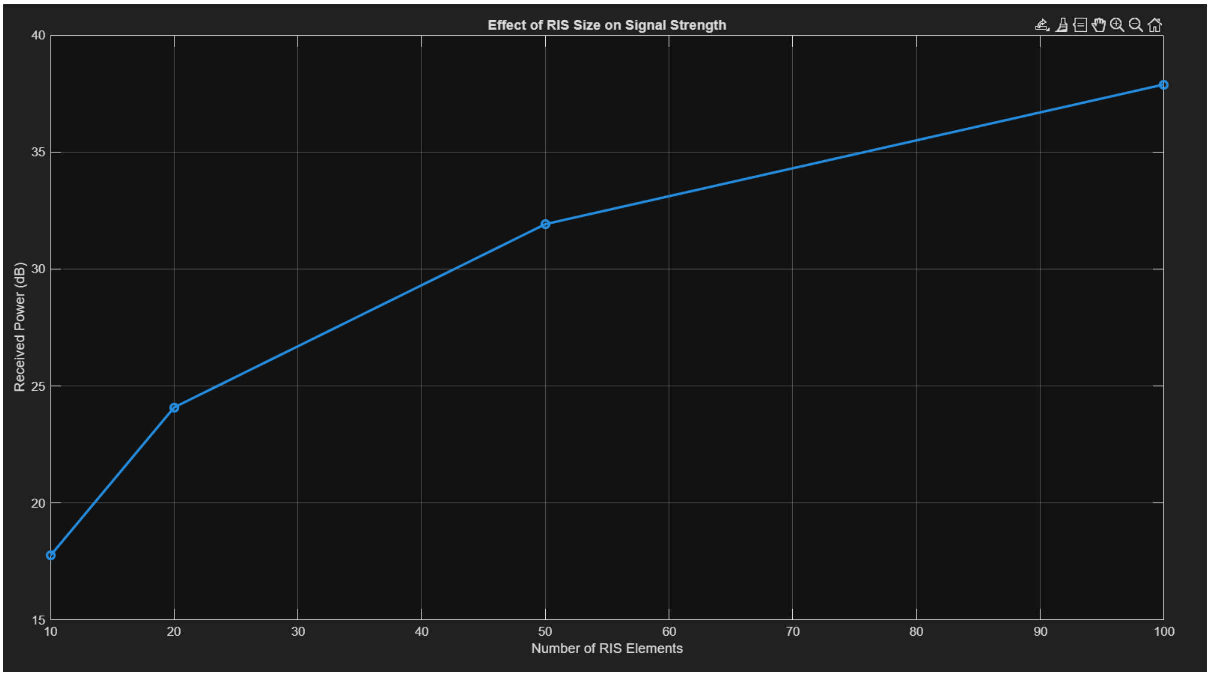

Figure 2: Received Power vs Number of RIS Elements under optimized phase configuration (200 Monte Carlo iterations)

Figure 2 illustrates how received signal power grows as the number of RIS elements increases from N=10 to N=100. At N=10, the received power is approximately 18 dB, which rises to nearly 38 dB at N=100. The curve follows a logarithmic growth pattern, consistent with the theoretical expectation that received power scales as N^2 under optimized phase alignment, which appears linear in the dB scale. This confirms that deploying more RIS elements yields significant beamforming gain, though with diminishing returns at larger N values.

Gain Achieved Using RIS Phase Optimization

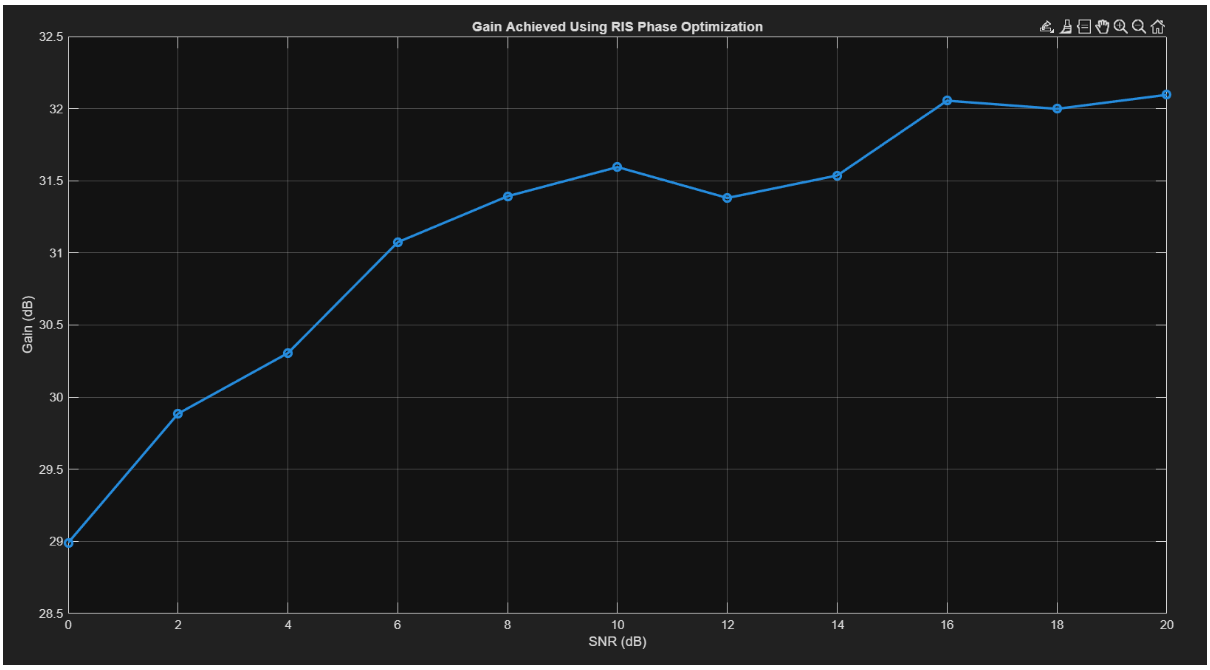

Figure 3: Gain (dB) of Optimized RIS over No-RIS baseline vs SNR

Figure 3 shows the gain achieved by optimized RIS over the no-RIS baseline across SNR values. At 0 dB SNR, the gain is approximately 29 dB, and it increases steadily to around 32 dB at higher SNR values. The rising trend occurs because as SNR increases, the noise contribution to the no-RIS scenario diminishes, making the optimized RIS advantage more pronounced. Minor fluctuations in the curve are attributed to Monte Carlo randomness over 500 iterations. Overall, the consistently high gain across all SNR levels confirms the robustness of RIS phase optimization under varying noise conditions.

Phase Distribution: Random vs Optimized

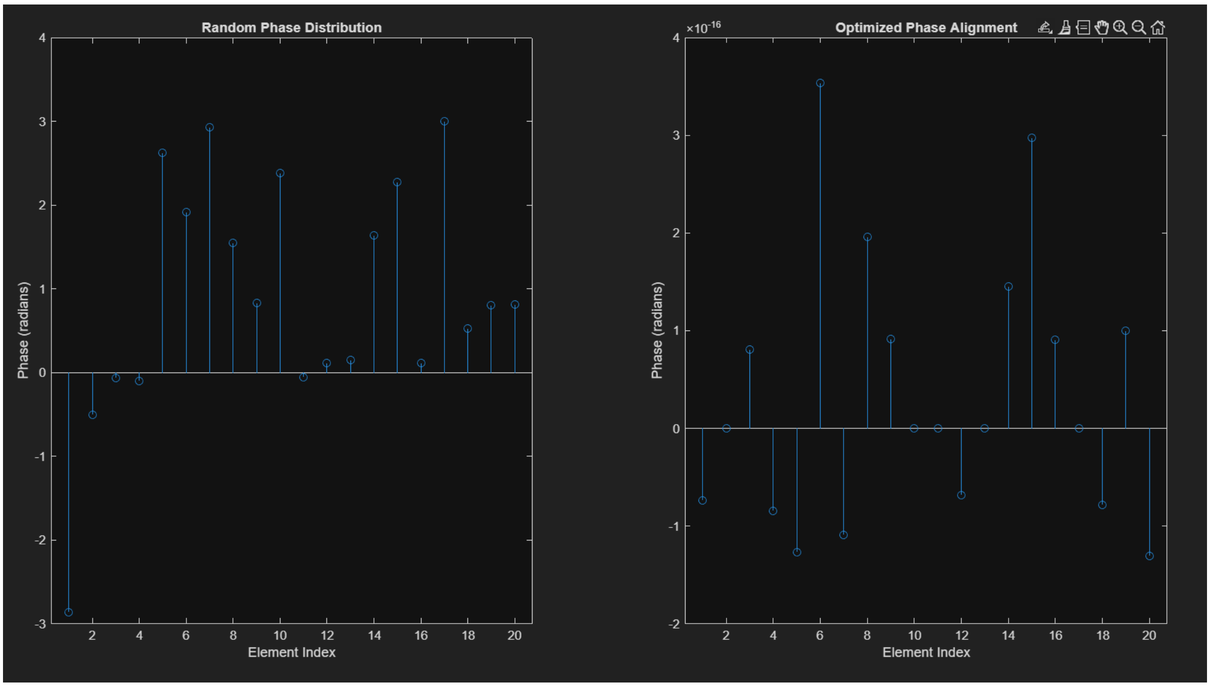

Figure 4: Phase distribution of 20 RIS element contributions: Random phase (left) vs Optimized phase alignment (right)

Figure 4 visually demonstrates the effect of phase optimization across 20 RIS elements. The left subplot shows the random phase case, where element phases are scattered between $-\pi$ and π, resulting in both constructive and destructive interference at the receiver. The right subplot shows the optimized case, where the residual phase of each element's contribution is reduced to values on the order of 10^{-16} radians — effectively zero — confirming that all reflected signals arrive perfectly in phase. This near-zero phase alignment is the direct result of applying $\theta_{opt} = -\angle(h_r \odot g_r)$, which cancels the channel phase entirely and maximizes constructive interference at the receiver.

Conclusion

This paper presented a MATLAB-based simulation study of Reconfigurable Intelligent Surfaces in wireless communication systems. Three scenarios were evaluated: without RIS, with randomly configured RIS, and with optimized RIS, across SNR values ranging from 0 to 20 dB using 500 Monte Carlo iterations.

The results demonstrated that optimized RIS phase configuration consistently achieved approximately 32 dB of received power across all SNR levels, significantly outperforming both the random RIS case at 17 dB and the no-RIS baseline which degraded toward 0 dB at high SNR. The gain achieved by optimized RIS over the no-RIS baseline ranged from 29 dB to 32 dB, confirming the effectiveness of phase optimization under varying noise conditions.

The analysis of RIS array size showed that increasing the number of elements from N=10 to N=100 raised the received power from 18 dB to 38 dB, following a logarithmic growth pattern consistent with the theoretical N^2 power scaling law. This confirms that larger RIS arrays provide substantial beamforming gain, though with diminishing returns at higher element counts.

The phase distribution analysis visually confirmed that optimized phase alignment reduces the residual phase of all reflected signals to effectively zero, converting destructive interference into constructive interference. In contrast, random phase assignment resulted in scattered phases between $-\pi$ and π, yielding inconsistent and suboptimal signal combining.

Overall, the simulation validates that RIS with optimized phase control is a highly effective technique for improving received signal strength in wireless systems, with practical benefits scaling directly with the number of deployed elements.

MATLAB Code

```
clc;
clear;
close all;

N = 50;
SNR_dB = 0:2:20;
num_iter = 500;

snr_noRIS = zeros(size(SNR_dB));
snr_randRIS = zeros(size(SNR_dB));
snr_optRIS = zeros(size(SNR_dB));

for i = 1:length(SNR_dB)

   p1 = 0; p2 = 0; p3 = 0;

   for k = 1:num_iter

      % Channels
      h_d = (randn + 1j*randn)/sqrt(2);
      h_r = (randn(N,1) + 1j*randn(N,1))/sqrt(2);
      g_r = (randn(N,1) + 1j*randn(N,1))/sqrt(2);

      % Noise based on SNR
      noise_power = 10^(-SNR_dB(i)/10);
      noise = sqrt(noise_power/2)*(randn + 1j*randn);

      % WITHOUT RIS
      p1 = p1 + abs(h_d + noise)^2;

      % RANDOM RIS
      theta_rand = exp(1j*2*pi*rand(N,1));
      ris_rand = sum(h_r .* g_r .* theta_rand);
      p2 = p2 + abs(h_d + ris_rand + noise)^2;

      % OPTIMIZED RIS
      theta_opt = exp(-1j*angle(h_r .* g_r));
      ris_opt = sum(h_r .* g_r .* theta_opt);
      p3 = p3 + abs(h_d + ris_opt + noise)^2;

   end

   snr_noRIS(i) = p1/num_iter;
   snr_randRIS(i) = p2/num_iter;
   snr_optRIS(i) = p3/num_iter;
end

figure;
plot(SNR_dB, 10*log10(snr_noRIS), '-o','LineWidth',2); hold on;
plot(SNR_dB, 10*log10(snr_randRIS), '-s','LineWidth',2);
plot(SNR_dB, 10*log10(snr_optRIS), '-^','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Received Power (dB)');
title('SNR Comparison: RIS vs No RIS');
legend('Without RIS','Random RIS','Optimized RIS');
```

```
N_values = [10 20 50 100];
power_vs_N = zeros(size(N_values));
num_iter2 = 200;

for i = 1:length(N_values)

    N_temp = N_values(i);
    temp_power = 0;

    for k = 1:num_iter2

        h_r = (randn(N_temp,1)+1j*randn(N_temp,1))/sqrt(2);
        g_r = (randn(N_temp,1)+1j*randn(N_temp,1))/sqrt(2);

        theta_opt = exp(-1j*angle(h_r .* g_r));
        ris_effect = sum(h_r .* g_r .* theta_opt);

        temp_power = temp_power + abs(ris_effect)^2;
    end

    power_vs_N(i) = temp_power / num_iter2;
end

figure;
plot(N_values, 10*log10(power_vs_N), '-o','LineWidth',2);
grid on;
xlabel('Number of RIS Elements');
ylabel('Received Power (dB)');
title('Effect of RIS Size on Signal Strength');

gain_dB = 10*log10(snr_optRIS) - 10*log10(snr_noRIS);

figure;
plot(SNR_dB, gain_dB, '-o','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Gain (dB)');
title('Gain Achieved Using RIS Phase Optimization');

N_phase = 20;

h_r = (randn(N_phase,1)+1j*randn(N_phase,1))/sqrt(2);
g_r = (randn(N_phase,1)+1j*randn(N_phase,1))/sqrt(2);

% Random phase
theta_rand = exp(1j*2*pi*rand(N_phase,1));
signal_rand = h_r .* g_r .* theta_rand;

% Optimized phase
theta_opt = exp(-1j*angle(h_r .* g_r));
signal_opt = h_r .* g_r .* theta_opt;

figure;

subplot(1,2,1);
stem(angle(signal_rand));
title('Random Phase Distribution');
xlabel('Element Index'); ylabel('Phase (radians)');
```

```
subplot(1,2,2);
stem(angle(signal_opt));
title('Optimized Phase Alignment');
xlabel('Element Index'); ylabel('Phase (radians)');
```